INDUSTRIAL TRAINING PRESENTATION (ITR) · 2026

Amit Astro

Intelligent Vedic Astrology & Real-Time Astrological Consultation Platform.

SUBMITTED BY

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SUBMITTED TO

[Faculty Name]

Department of Computer Science & Engineering

ENGINE SPECS: REACT 18 & TYPESCRIPT · NODE.JS & EXPRESS · POSTGRESQL & SQLITE · SOCKET.IO LIVE CHAT · GEMINI AI VEDIC ENGINE · RAZORPAY BILLING

ARCHITECTURE & INDEX

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ORGANIZATIONAL PROFILE

Company Introduction

Industry host organization supporting the full-stack software training tenure.

ESTABLISHED [YEAR] HQ: [CITY, STATE, COUNTRY]

[Organisation Name]

[Organisation Name] is a modern software engineering and digital product innovation enterprise specializing in [domains / services, e.g. High-Scale Web Platforms, Distributed Event Streaming, and Applied AI Integrations].

CORE DOMAINS

Full-Stack Cloud Architecture, Real-Time WebSockets & AI Systems

CLIENT INDUSTRIES

Serving consumer tech, real-time consulting, and astrological analytics

📍 **Office Address:** [Office address, Tech Park / Cyber City Hub]



Company Logo

Paste or drag company logo here for formal presentation review

FORMAL VALIDATION

Training Certificate

Official validation of industrial internship completion and software engineering delivery.

Paste Training / Internship Certificate Here

Embed high-resolution scanned copy of the official training completion certificate signed by the engineering mentor and authorized authority.

VALIDATION CRITERIA

- ✓ Tenure Hours Successfully Completed
- ✓ Kundli & Real-Time Modules Deployed
- ✓ GitHub Git Hygiene & Code Reviews
- ✓ Faculty Assessment Ready

CANDIDATE DETAILS

Mahansh Gaur

Project: Amit Astro (Kundli Platform)

VERIFIED ITR CANDIDATE

TECHNICAL ECOSYSTEM

Technology Stack Used

Modern, typed, and enterprise-ready technologies powering the frontend, backend, and real-time socket layer.

React 18 & TS

Declarative component hierarchy, strict TypeScript type checking, custom hooks, and Vite-powered compilation.

Node & Express

High-concurrency asynchronous REST backend handling business controllers, birth profiles, and secure routing.

PostgreSQL & SQLite

Dual-database abstraction with connection pooling for production Render deployments and local zero-setup dev.

Socket.IO

Full-duplex real-time WebSockets facilitating instant live astrologer consultation chat, typing states, and notifications.

Google Gemini AI

Bilingual LLM pipeline grounded in classical Parashari shastras for horoscope interpretations and remedies.

Razorpay Gateway

Payment intents, recurring subscription billing (Family 360), and HMAC-SHA256 verified webhook processing.

Tailwind CSS

Responsive design tokens, glassmorphic UI cards, Lucide icon system, and seamless light/dark layout structures.

Cloudflare & Render

Continuous integration and edge deployment: Cloudflare Pages for client and Render cloud services for backend.

LEARNING OBJECTIVES

Goal of Industrial Training

Five core full-stack software and computational competencies targeted and mastered during the internship.

01 **Full-Stack Cloud Architecture**
Architect an end-to-end distributed system: React client ↔ Express REST APIs ↔ WebSocket server ↔ PostgreSQL database.
MASTERED

02 **Computational Ephemeris & Kundli Algorithms**
Translate astronomical ephemeris equations, Lahiri Ayanamsha, and house cusp algorithms into high-precision SVG chart generators.
MASTERED

03 **Domain-Specific LLM Engineering**
Ground Google Gemini AI prompts in Brihat Parashara Hora Shastra rules for accurate, bilingual (Hindi & English) horoscopes.
MASTERED

04 **Real-Time Bidirectional Event Streaming**
Implement robust Socket.IO pipelines with typing indicators, presence tracking, message delivery acknowledgments, and stars.
MASTERED

05 **Security, Payments & Commercial Hygiene**
Implement RSA encrypted payloads, CSRF double-cookie tokens, Razorpay subscriptions (Family 360), and Git CI/CD deployments.
MASTERED

PRODUCT VISION

Amit Astro (Kundli Engine) Overview

Bridging classical astronomical science with modern cloud engineering and artificial intelligence.

"Amit Astro turns birth coordinates and timestamps into sub-second mathematical ephemeris calculations, predictive dasha timelines, and real-time live consultations."

The platform ingests exact birth date, time, and geographical coordinates, computes planetary sidereal longitudes with Lahiri Ayanamsha, renders North/South Indian Kundli charts (D1 to D60), calculates Vimshottari Dashas, and powers an interactive astrologer-seeker ecosystem with bilingual Gemini AI analysis and live Socket.IO messaging.



01 · INPUT

Birth Coordinates

Date of birth, exact time (UTC/IST), city geocoding, latitude & longitude



02 · PROCESS

Ephemeris → Chart → Dasha → AI

Planetary longitudes, house cusps, 120-yr Vimshottari dasha, and Gemini AI synthesis



03 · OUTPUT

Kundli Charts & Live Chat

Dynamic SVG charts, Gochar transits, AI reports, and live consultations

MARKET NEED & INNOVATION

Problem Statement & Proposed Solution

Modernizing astrological consultations by eliminating inaccurate ephemeris engines and unsecure chats.

The Legacy Problem

FRICTION POINTS

- **Inaccurate Calculations & Lag**
  Legacy apps use slow web scrapers or approximate formulas that miscalculate Lagna and Bhav Chalit cusps.
- **Fragmented Unsecure Consultations**
  Seekers and astrologers communicate via WhatsApp without birth chart context, audit records, or verified timings.
- **Generic, Cookie-Cutter Horoscopes**
  Pre-written static paragraphs with no personalized Parashari shastra depth or bilingual remedies.
- **Zero Multi-Profile Family Architecture**
  No ability to save multiple family charts under one account or conduct Gun Milan matching seamlessly.

The Amit Astro Solution

ENGINEERED FIX

- **Sub-Second Mathematical Ephemeris**
  Precision astronomical calculation of planetary longitudes, retrogrades, combustions, and D1-D60 charts.
- **Integrated Socket.IO Consultation**
  Real-time consultation chat with live birth-chart sidebar, 5-minute free trials, and verified timers.
- **Bilingual Gemini AI Vedic Engine**
  Domain-grounded AI generates Hindi & English predictions, Raj Yogas, and classical Parashari remedies.
- **Family 360 & Razorpay Subscriptions**
  Multi-profile vault (self, spouse, children, parents), instant Gun Milan matching, and automated billing.

TECHNICAL BLUEPRINT

System Architecture

A high-throughput, microservice-ready full stack connecting mathematical computation to live sockets.

DONE	DONE	DONE	DONE	DONE	DONE	DONE
01 · Seeker UI	**02 · Ephemeris**	**03 · Chart Engine**	**04 · Dasha/Transit**	**05 · Gemini AI**	**06 · Live Socket**	**07 · CRM & Pay**
React 18 / Vite	Lahiri Math	SVG D1-D60	Vimshottari	Vedic Insights	Socket.IO Chat	Razorpay & DB

Mathematical Rigor

Sub-second astronomical ephemeris calculations ensure exact planetary degrees, retrograde states, combustion, and divisional vargas without third-party API dependencies.

Sub-50ms Reactivity

Full-duplex WebSocket connections deliver instantaneous astrologer chat messages, typing events, read receipts, and consultation countdowns seamlessly.

Enterprise Security

Hardened with HTTP-only JWT cookies, CSRF protection tokens, RSA payload encryption, parameterized queries, and Razorpay webhook HMAC validation.

CODEBASE LAYOUT

Working Project: Code Structure

Clean monorepo-style organization with strict separation of frontend UI, server controllers, and database services.

Kundli/ — monorepo tree

```
Kundli/
├── client/                # React 18 + Vite Frontend
│   ├── src/pages/customer/    # Seeker Portal & Kundli charts
│   ├── src/pages/admin/       # CRM & Live Astrologer console
│   └── src/utils/api.ts       # RSA-encrypted API client
└── server/                # Express + TypeScript Backend
    ├── src/controllers/       # Auth, Chat, Payment, Trial
    ├── src/services/          # Gemini AI, Subscriptions, Security
    ├── src/db/database.ts     # Postgres & SQLite schema manager
    └── src/index.ts           # Express app + Socket.IO server
```

server/src/controllers/
Handles HTTP routes: authController, chatController, paymentController, birthProfileController, and adminController.

server/src/services/geminiService.ts
Encapsulates Google Gemini AI SDK, handling Parashari astrological prompt engineering and bilingual Hindi/English synthesis.

server/src/db/database.ts
Enterprise database layer supporting both Render PostgreSQL pool and local SQLite3 with automatic migrations.

client/src/pages/customer/
Seeker dashboard: North/South Indian Kundli SVG canvas, Mahadasha timeline, Panchang, and real-time live chat.

client/src/pages/admin/
Comprehensive CRM: real-time consultation desk, customer lifecycle, 5-minute trial grants, Family 360 badges, and revenue analytics.

● CORE ENGINE · PART 1

Working Project: Ephemeris & Kundli Engine

Precision sidereal calculations, Lahiri Ayanamsha calibration, and dynamic vector chart generation.

zsh — npm run test:ephemeris

```
[EPHEMERIS] Ingesting Target: 19/01/2005 12:00 PM | Udaipur, Rajasthan (24.5854° N, 73.7125° E)
[COMPUTED]  Lagna: Aries (Mesha) at 14°28' | Moon: Taurus (Rohini Nakshatra, Pada 2)
[CHARTS]    Generated D1 (Rashi), D9 (Navamsha), D10 (Dashamsha), Bhav Chalit in 18ms
```

16

PLANETS & UPAGRAHAS

Sun, Moon, Mars, Rahu, Ketu, Gulika & Mandi

16+

DIVISIONAL VARGAS

D1 Rashi, D9 Navamsha, D10 Career, D60 Shashtiamsha

< 0.01°

MATHEMATICAL PRECISION

Verified against Chitra Paksha (Lahiri) Ayanamsha



Dynamic SVG charts adapt instantly to both **North Indian Diamond** and **South Indian Box** styles, dynamically rendering retrograde tags, combustion markers, and house lord indicators.

CORE ENGINE · PART 2

Working Project: Dasha Engine & Gun Milan

120-year Vimshottari dasha hierarchy, planetary transits (Gochar), and 36-point Ashtakoot matching.

120 Yrs
VIMSHOTTARI LIFECYCLE
Continuous calculation from birth Nakshatra balance

3 Tiers
DASHA HIERARCHY
Mahadasha → Antardasha → Pratyantardasha

36 / 36
GUN MILAN MATCHING
Varna, Vashya, Tara, Yoni, Maitri, Gana, Bhakoot, Nadi

PREDICTIVE ASTROLOGICAL PIPELINE

BIRTH NAKSHATRA & LONGITUDE

```
Moon: Rohini Nakshatra (46.8°) → Moon Dasha balance: 4y 2m 11d
```

→

HIERARCHICAL TIMELINE OBJECT

```
{{"mahadasha": "Mars", "start": "2009-03-30", "antardasha": "Jupiter", "sub": "Saturn"}}
```

Feeds active Gochar (planetary transits) over natal houses, identifying Sade Sati, Dhaiya, and planetary returns.

AI INNOVATION

Working Project: AI Vedic Consultation Engine

Domain-grounded Google Gemini AI delivering personalized, bilingual (English & Hindi) Parashari readings.

VEDIC GROUNDING

Classical Rules

Parashari Shastra Knowledge

Prompts are grounded in classical texts (Brihat Parashara Hora Shastra, Phaladeepika):

- Gajakesari Yoga (Jupiter in Kendra from Moon)
- Raj Yoga & Vipreet Raj Yoga configurations
- Sade Sati 7.5-year Saturn transit phases

📜 Eliminates generic AI hallucinations with strict astrological constraint rules.

BILINGUAL LLM

Gemini 1.5/2.0

Multilingual Synthesizer

Full dual-language output generation customized for modern seekers:

- Natural Devanagari Hindi (हिन्दी) & English generation
- Tailored practical remedies: Mantras, Gemstones & Fasting

⚡ Empowers seekers with actionable life, career, and marriage clarity.

Planetary Coordinates → Dasha & Yogas Extracted → Parashari Prompt Injected → Gemini LLM Inference → Bilingual Astrological Report

PLATFORM INTEGRATION

Working Project: Live Consultation, Billing & CRM

Unifying WebSocket real-time chat, Razorpay subscriptions, and high-efficiency administrative CRM.

LIVE SOCKET

Socket.IO Chat

Full-duplex real-time consultation desk:

- Real-time typing indicators & message read status
- Starred messages, reactions & message pinning
- Integrated birth-profile side drawer for the astrologer
- Automated 5-minute trial countdown locks

PAYMENTS

Razorpay & Plans

Tiered subscription monetization:

- **Family 360 Plan:** Up to 4 family birth charts
- **Pro & Lite:** Single chart + priority consults
- Instant webhook verification via HMAC-SHA256
- Automated invoice generation and ledger entries

ADMIN CRM

Astrologer CRM

Enterprise administrative control:

- Complete Seeker Directory (13+ live users)
- 1-Click Trial Grants & Family 360 manual overrides
- Astrologer private consultation notes & tags
- Real-time financial revenue metrics (₹5,897+)

MILESTONE TRACKER

Development Progress

Current execution status across project milestones deployed to Cloudflare Pages & Render cloud.

01 DONE

Architecture & Dual DB

Postgres on Render + SQLite local, JWT session security.

02 DONE

Ephemeris & Kundli Engine

16 planets, D1-D60 SVG charts, Lahiri Ayanamsha math.

03 DONE

Dasha & Gun Milan

120-year Vimshottari hierarchy + 36-point compatibility.

04 DONE

Gemini AI Vedic Engine

Parashari-grounded bilingual (Hindi/English) interpretations.

05 DONE

Socket.IO Live Chat

Real-time bidirectional consultations with 5-min trial locks.

06 DONE

Billing, CRM & Deploy

Razorpay Family 360, admin controls & live cloud release.

🎯 **Live Production App:** amitastro.pages.dev

💻 Code repository: **github.com/mahanshstarktech/amitastro**

TECHNICAL PROBLEM SOLVING

Engineering Challenges & Solutions

Key obstacles encountered during the project implementation and the robust architectural solutions applied.

PostgreSQL vs SQLite Identifier Variance	Postgres treated double quotes as column identifiers causing runtime crashes; resolved by parameterizing all SQL literals.
Astronomical Ephemeris Coordinate Drift	Standard algorithms drifted over leap years; calibrated against Chitra Paksha Lahiri Ayanamsha for sub-0.01° precision.
WebSocket Mobile Drop-off Reconnection	Mobile browser tab suspensions terminated chats; implemented heartbeat keep-alives and state rehydration on reconnect.
Payment Webhook Integrity & Tampering	Prevented subscription spoofing using cryptographically signed HMAC-SHA256 signature verification on Razorpay events.
AI Hallucinations in Astrological Readings	Grounded LLM prompts with Parashari shastra rulebooks and structured JSON schemas to ensure canonical remedies.

VISION & SCALABILITY

Future Scope & Commercial Roadmap

Strategic features to expand Amit Astro into an international enterprise astrology ecosystem.

Native iOS & Android Mobile Apps

Build high-performance native apps via React Native with push notifications for Gochar planetary transits and daily Panchang.

1-on-1 WebRTC Video Consultations

End-to-end encrypted audio/video consultation rooms with live screen-sharing of birth charts between astrologer and client.

Automated 50-Page Kundli Book

Server-side headless Chromium PDF generation compiling full D1-D60 charts, 120-year Dasha tables, and gemstone remedies into a keepsake book.

Voice-Interactive AI Astrologer

Real-time conversational speech-to-speech AI consultation answering queries in Hindi, Sanskrit, and English.

Global Multi-Currency Stripe Billing

International gateway integration (Stripe, PayPal) supporting multi-currency pricing (USD, EUR, GBP, AED) for global diaspora seekers seeking Vedic consultation.

TECHNICAL DEFENSE

Questions & Discussion

Floor open for architectural inquiry, computational review, and faculty evaluation.

Ephemeris Mathematical Accuracy

Socket.IO Latency & Concurrency

Gemini AI Shastra Grounding

CONCLUSION

Thank You

Mahansh Gaur

github.com/mahanshstarktech/amitastro

Amit Astro — Intelligent Vedic Astrology & Real-Time Consultation Platform